Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	*(a)*	(i)		rate = $k[CH_3I][S_2O_3^{2-}]$ award (1) for rate equation containing $k[CH_3I]$ award (1) for second order rate equation		 1	 1	 2		
		(ii)		change E_a to 81100 J mol^{-1} (1) rearrange equation to $T = -\frac{Ea}{\left(R\ ln\frac{k}{A}\right)}$ (1) T = 346 K (1)	1		 1 1	 3	 3	
	(b)	(i)		quenching is the sudden stopping / significant slowing of a chemical reaction in a sample (1) ensure sample composition does not change between taking sample and analysis OR during analysis (1)	1 1			 2		 2
		(ii)		starch	1			1		1
		(iii)		canary/bright yellow	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	I	catalyst in the same physical state as reaction	1			1		
			II	award (1) each for any **two** of following • transition metals have empty orbitals so can form bonds to reactant molecules / form complex • transition metals have variable oxidation states (so they can oxidise/reduce the reactants) • products are released and the metal returns to the original oxidation state	2			2		
		(ii)		$[H^+]^2 = K_a \times [acid]$ $[H^+] = 1.97 \times 10^{-3}$ mol dm^{-3} pH = 2.7		3		3	3	
				Question 10 total	**8**	**4**	**3**	**15**	**6**	**4**